Practice Set 10 Paper 2 – Urology

Contents

1. Q1. Describe staging of CA prostate and management of non-organ confirmed CA prostate (LAQ)
2. Q2. Classify CA bladder. Discuss in details organ preserving surgeries for TCC of bladder. (LAQ)
3. Q3a. Allopurinol (SAQ)
4. Q3b. Wilm's tumor (SAQ)
5. Q3c. Hemodialysis access. (SAQ)
6. Q3d. TRUS (SAQ)
7. Q3e. Von Bruns Nest (SAQ)
8. Q3f. Primary Hyperoxaluria (SAQ)

Practice paper compiled from the question bank.

Q1. Describe staging of CA prostate and management of non-organ confirmed CA prostate

Type: LAQ (25 marks)

Read as: staging of prostate cancer, and management of non-organ-confined (locally advanced and node-positive, non-metastatic) disease.

Introduction

- At least 10% of new prostate cancers are locally advanced (T3–4, N0 or N+, M0). Any single treatment carries a high recurrence risk, so **multimodal therapy** is the rule.

Staging: TNM (tumour, node, metastasis) 2017 (AJCC 8th edition)

Stage	Definition
T1a / T1b	Incidental at TURP (transurethral resection of prostate): ≤5% / >5% of tissue
T1c	Not palpable; found on needle biopsy (for raised PSA)
T2a / T2b / T2c	Palpable, confined: ≤ half of one lobe / > half of one lobe / both lobes
T3a	Extraprostatic extension (unilateral or bilateral)
T3b	**Seminal vesicle invasion** (not T4)
T4	Fixed or invading sphincter, rectum, bladder, levators, pelvic wall
N1	Regional (pelvic) nodes
M1a / M1b / M1c	Non-regional nodes / bone / other sites ± bone

- **Prognostic stage groups** add PSA and grade group: stage III = T3–4, PSA ≥20 ng/mL or grade group 5; IVA = N1; IVB = M1. pT2 is not subdivided.
- **Grade group (ISUP, International Society of Urological Pathology):** GG1 ≤6; GG2 3+4; GG3 4+3; GG4 8; GG5 9–10.

Risk groups

System	High risk	Very high / locally advanced
D'Amico	T2c, PSA >20 or Gleason 8–10 (any one)	–
NCCN (National Comprehensive Cancer Network)	cT3a, GG4–5 or PSA >20	cT3b–T4, primary Gleason pattern 5, ≥2 high-risk features, or >4 cores GG4–5
EAU (European Association of Urology)	PSA >20, ISUP 4–5 or cT2c	**Locally advanced: cT3–4 or cN+, any PSA, any grade**

- **Also know:** CAPRA (Cancer of the Prostate Risk Assessment) score; Partin tables; Kattan nomograms; Decipher.

Staging work-up

- **DRE (digital rectal examination)** sets clinical T stage. PSA (prostate-specific antigen) >20 ng/mL: nodes in about 20%.
- **mpMRI (multiparametric MRI):** separates T2 from T3 (PI-RADS v2 reporting); best in higher-risk men.
- **Bone scan + CT/MRI** for high risk, PSA >20 or GG4–5.
- **PSMA (prostate-specific membrane antigen) PET-CT:** proPSMA trial: accuracy 92% vs 65% for conventional imaging.

Management of locally advanced N0 (cT3–4 N0 M0)

A. Radiotherapy (RT) + long-term ADT (androgen deprivation therapy): standard

- Dose-escalated IMRT (intensity-modulated RT), or EBRT (external-beam RT) with brachytherapy boost (ASCENDE-RT).
- **Why add ADT:** it shrinks the target and acts synergistically with radiation.
- **EORTC 22863:** 3 years of goserelin with RT improved 5-year OS (overall survival), 78% vs 62%.
- **RTOG 92-02:** 28 vs 4 months ADT better; OS gain in Gleason 8–10. NCCN: ADT 18–36 months.
- **RT is still needed with ADT:** SPCG-7 (cancer death RR (relative risk) 0.44) and MRC PR07 (OS HR (hazard ratio) 0.77) showed ADT alone is inferior.
- **STAMPEDE (abiraterone, M0):** for N1, or two of T3–4, Gleason ≥8 or PSA ≥40, **abiraterone + prednisolone for 2 years** with ADT + RT gave metastasis-free survival HR 0.53 and OS HR 0.60. NCCN now lists it for very high risk.
- Pelvic nodal RT: RTOG 94-13 inconclusive; POP-RT (Tata Memorial) improved biochemical failure-free survival in high-risk N0.

B. Radical prostatectomy (RP) + extended PLND (pelvic lymph node dissection), within multimodal therapy

- For fit men with a resectable tumour (mainly cT3a). cT3 is overstaged in 7–27%.
- Neoadjuvant ADT before RP gives **no survival gain**.
- **Adjuvant vs early salvage RT:** RADICALS-RT, RAVES and GETUG-AFU 17 favour **early salvage** for most men. Adjuvant RT is still considered for very high risk (pT3b, GG4–5, positive margins).

C. ADT alone: only for men unfit for or refusing local treatment. Consider immediate ADT if PSA doubling time is short or PSA is very high.

Node-positive disease

Setting	Management
cN1 M0	RT to prostate and pelvis + ADT 2–3 years + abiraterone 2 years (STAMPEDE). RP + extended PLND in selected men
pN1 after RP	**Immediate ADT** (ECOG 7887: OS benefit) **or ADT + adjuvant RT** (better survival than ADT alone); observation with undetectable PSA and ≤2 nodes

- **Also know:** RTOG 0521 docetaxel + RT (OS HR 0.69, not routine); cryotherapy and HIFU (high-intensity focused ultrasound) not standard.

Algorithm

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cT3–4 / cN1, M0 (PSMA PET or CT + bone scan)
   ├─ Fit, resectable → RP + PLND → early salvage RT ± ADT
   ├─ Most → RT + ADT 18–36 months (+ abiraterone if N1/very high risk)
   └─ Unfit → ADT alone
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Recent advances

- PSMA PET staging; trials adding apalutamide or enzalutamide (ATLAS, ENZARAD); SPCG-15 (RP vs RT + ADT).

Diagram to draw: TNM T-stage diagram (T2 confined, T3a extracapsular, T3b seminal vesicle, T4 adjacent organs).

Key points to remember:

- Seminal vesicle invasion is T3b; non-regional nodes are M1a.
- RT + long-term ADT (18–36 months) is the standard; ADT alone is inferior (SPCG-7).
- STAMPEDE: abiraterone 2 years for N1 or very high-risk M0 (metastasis-free survival HR 0.53).
- Early salvage RT preferred over adjuvant RT for most men after RP.

Q2. Classify CA bladder. Discuss in details organ preserving surgeries for TCC of bladder.

Type: LAQ (25 marks) (also answered in 2019-Summer Paper 4, Q2)

Read as: classification of bladder cancer, and bladder-preserving treatment of TCC (transitional cell carcinoma, urothelial carcinoma).

A. Classification

1. Histological

Type	Share	Note
Urothelial carcinoma	~90%	Variants: micropapillary, plasmacytoid, sarcomatoid, nested; squamous or glandular differentiation
Squamous cell carcinoma	~5%	Schistosomiasis, catheter, stones
Adenocarcinoma	~1–2%	Urachal (dome), exstrophy
Small cell / others	<1%	Neuroendocrine; sarcoma

2. Grade (WHO 2004/2016, World Health Organization)

- PUNLMP (papillary urothelial neoplasm of low malignant potential), low-grade and high-grade carcinoma. WHO 1973 (G1–G3) is still reported by some.
- Low-grade (FGFR3 pathway) recurs but rarely progresses. High-grade and CIS (carcinoma in situ; TP53/RB pathway) progress.

3. Stage: TNM 2017

T	Definition
Ta / Tis	Non-invasive papillary / flat CIS
T1	Lamina propria
T2a / T2b	Inner / outer half of muscle
T3a / T3b	Perivesical fat, microscopic / macroscopic
T4a / T4b	Prostatic stroma, uterus, vagina / pelvic or abdominal wall

- N1 one true-pelvic node; N2 ≥2 pelvic nodes; N3 common iliac; M1a distant nodes; M1b organs.

4. Clinical groups

- **NMIBC (non-muscle-invasive, ~70%)**: Ta, Tis, T1. Risk groups: low (solitary primary low-grade Ta <3 cm), intermediate, high (high-grade, T1 or CIS).
- **MIBC (muscle-invasive)**: T2–T4a. Metastatic.
- **Molecular:** luminal papillary versus basal/squamous subtypes.

B. Organ-preserving surgery

Why: RC (radical cystectomy) is morbid and patients are elderly. Single modalities fail because they leave either the remaining urothelium or occult nodes untreated (cT2 understaged to pT3 in up to about 40%, occult nodes about 25%).

1. TURBT (transurethral resection of bladder tumour) for NMIBC: the standard

- Complete resection with **detrusor muscle in the specimen**; separate base biopsies; cold-cup biopsy of suspicious areas.

- **En bloc resection** (loop or laser) keeps architecture, gives more detrusor and fewer perforations.
- **Repeat TURBT within 6 weeks** for all T1, incomplete resection or no muscle. Residual tumour in ≥40%.
- Then intravesical chemotherapy or BCG (bacillus Calmette–Guérin) by risk group.

2. Radical TURBT for MIBC

- Only for tumour in superficial muscle with a negative re-resection; about 10% are pT0 at cystectomy.
- About a third later develop MIBC. Not a primary treatment.

3. Partial cystectomy

Ideal candidate	Contraindication
Solitary tumour **<3 cm** in a mobile part (dome)	**CIS**, multifocal tumour
No hydronephrosis	Trigone or bladder neck tumour
1–2 cm margin possible; adequate capacity left	Small bladder; prostatic urethral involvement
Urachal adenocarcinoma (treatment of choice)	

Steps: NAC (neoadjuvant cisplatin chemotherapy) if eligible → PLND (pelvic lymph node dissection) → cystotomy away from tumour → full-thickness excision with perivesical fat → **frozen-section margins** → ureteric reimplantation if needed → watertight closure → drain. For urachal cancer, en bloc dome, urachus and umbilicus.

Outcomes: 5-year OS (overall survival) 67–70%; intact bladder 65–74%; salvage RC up to a quarter. Late recurrence, so lifelong follow-up.

4. TMT (trimodality therapy): the best-studied option

- **Components:** maximal visibly complete TURBT + RT (radiotherapy) + concurrent radiosensitising chemotherapy.
- **Selection:** cT2, solitary, **no hydronephrosis, no extensive CIS**, complete TURBT, good bladder function, compliant patient. Only 6–19% of MIBC qualify.

Radiosensitiser	Evidence
Mitomycin C + 5-FU (fluorouracil)	**BC2001:** 2-year locoregional DFS (disease-free survival) 67% vs 48% for RT alone
Cisplatin-based	MGH (Massachusetts General Hospital)/RTOG (Radiation Therapy Oncology Group) protocols
Gemcitabine; carbogen + nicotinamide	Cisplatin-ineligible; hypoxia modification

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TURBT → chemoRT 64 Gy/32 fr or 55 Gy/20 fr → cystoscopy + biopsy at 3–4 months
     ├─ Complete response → surveillance (NMIBC recurrence: TURBT + BCG)
     └─ Persistent / recurrent MIBC → salvage cystectomy (10–30%)
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- **Outcomes:** 5-year CSS (cancer-specific survival) 50–84%. MGH/RTOG: complete response 69%; 80% of 5-year survivors keep the bladder. Propensity-matched RC vs TMT: similar CSS (81% vs 84%).

Comparison

	Radical cystectomy	Bladder preservation
Staging	Pathological	Clinical only
Function	Diversion, sexual dysfunction	Native bladder; RT toxicity
Follow-up	Upper tract, urethra	**Cystoscopy 3-monthly for 2 years, to 10 years**, plus CT

Recent advances

- Checkpoint inhibitors with chemoradiation (trials ongoing); bladder sparing after complete response to NAC (RETAIN: about half recurred); robotic partial cystectomy.
- **Also know:** RT alone is quasi-palliative; chemotherapy alone not recommended.

Diagram to draw: TNM T-stage diagram of the bladder wall with a TMT flowchart.

Key points to remember:

- WHO 2004/2016: PUNLMP, low grade, high grade; T3b = macroscopic perivesical fat.
- Re-TURBT within 6 weeks for T1, no muscle or incomplete resection.
- Partial cystectomy: solitary <3 cm, no CIS, frozen-section margins, always PLND.
- TMT: BC2001 MMC/5-FU 67% vs 48%; salvage cystectomy 10–30%.

Q3a. Allopurinol

Type: SAQ (10 marks)

Introduction and mechanism

- A **purine (hypoxanthine) analogue** that inhibits **xanthine oxidase (xanthine dehydrogenase)**. This blocks hypoxanthine → xanthine → uric acid, lowering serum and urinary uric acid.
- Metabolised in the liver to **oxypurinol**, the active metabolite. It has a long half-life and is cleared by the kidney, so the dose is cut in renal impairment.

Uses in urology

Indication	Rationale
Recurrent calcium oxalate stones with hyperuricosuria (>600 mg/day) and normal urinary calcium	Urate "seeds" calcium oxalate (heterogeneous nucleation, salting out, inhibitor adsorption). **Ettinger (1986) RCT (randomised controlled trial): recurrence more than halved**; AUA (American Urological Association) recommends it
Uric acid stones	**Not first line:** low urine pH is the main cause, so alkalinise with potassium citrate to pH 6.0–6.5. Allopurinol is added for hyperuricosuria, gout or failure of alkalinisation
2,8-DHA (dihydroxyadenine) stones (APRT (adenine phosphoribosyltransferase) deficiency)	Blocks adenine → DHA; febuxostat more effective
Tumour lysis prophylaxis; gout; HPRT (hypoxanthine-guanine phosphoribosyltransferase) deficiency, PRPS (phosphoribosylpyrophosphate synthetase) overactivity	Prevents urate nephropathy

Dose

- Adult **300 mg/day** (start 100 mg, titrate); child 4–10 mg/kg/day (maximum 300 mg). Reduce for low creatinine clearance.

Adverse effects

- Rash (commonest), myalgia, gastrointestinal upset; gout flare on starting; interstitial nephritis.
- **Hypersensitivity in 2–8%**, from skin reactions to **allopurinol hypersensitivity syndrome** (fatal in up to 27%) and Stevens-Johnson syndrome. Risk is higher with high dose, renal failure, diuretics and statins. **HLA-B*58:01** carriers (some Asian populations) have higher risk.
- **Xanthine stones** (radiolucent): rare, mainly with high doses in Lesch-Nyhan syndrome.

Drug interactions

- **Azathioprine and 6-mercaptopurine:** xanthine oxidase metabolises them, so toxicity (marrow suppression) rises. Reduce their dose sharply (important after transplant).

Alternatives

- **Febuxostat:** non-purine xanthine oxidase inhibitor, no dose change in mild to moderate renal impairment. Carries a cardiovascular-death warning.
- **Also know:** rasburicase (urate oxidase) for tumour lysis.

Diagram to draw: purine pathway (hypoxanthine → xanthine → uric acid) showing the xanthine oxidase block.

Key points to remember:

- Xanthine oxidase inhibitor; active metabolite oxypurinol; 300 mg/day.
- Use: hyperuricosuric calcium oxalate stones (Ettinger 1986); uric acid stones need alkali first.
- Hypersensitivity 2–8%; HLA-B*58:01; cut azathioprine dose.

Q3b. Wilm's tumor

Type: SAQ (10 marks)

Read as: Wilms tumour (nephroblastoma).

Introduction

- WT (Wilms tumour): embryonal tumour of metanephric blastema; 95% of kidney cancers under 15 years; median age 3.5 years.

Genetics and syndromes

- Mostly sporadic; familial 1–2%. Genes: **WT1 (11p13)**, IGF2 (11p15), WTX, CTNNB1; **TP53 in anaplasia**.

Syndrome	WT risk
WAGR (Wilms, aniridia, genitourinary anomalies, intellectual disability; WT1 deletion)	50%
Denys-Drash (46,XY DSD (disorder of sex development), mesangial sclerosis; WT1 point mutation)	50%
BWS (Beckwith-Wiedemann syndrome), hemihypertrophy (11p15)	5–10%

- **Screening:** ultrasound 3–4-monthly to age 7 if risk >1%.
- **Nephrogenic rests** (precursors): over 90% of bilateral cases.

Pathology

- **Triphasic:** blastema, epithelium, stroma.
- **Anaplasia** (nuclei ≥3× larger, hyperchromasia, abnormal mitoses): unfavourable, chemoresistant; diffuse worse than focal.

Clinical features

- Abdominal mass, haematuria ~20%, hypertension ~25% (renin). Look for aniridia, hemihypertrophy, varicocele (IVC (inferior vena cava) thrombus); acquired von Willebrand disease.

Investigations

- Ultrasound with Doppler (IVC), contrast CT or MRI abdomen, **chest CT** (lung is the commonest metastatic site).

Staging (COG, Children's Oncology Group; surgical-pathological)

Stage	Description
I	Confined, completely resected, capsule intact
II	Beyond kidney (renal sinus vessels, perirenal fat), fully resected
III	Residual abdominal disease: **nodes, any spill**, positive margin, peritoneal implants
IV	Haematogenous metastases (lung, liver, bone, brain)
V	Bilateral

Treatment

- **COG:** upfront transperitoneal radical nephrectomy, avoid spill, **sample ≥5 lymph nodes**. **SIOP (International Society of Paediatric Oncology):** preoperative vincristine + dactinomycin, then surgery.

Group (favourable histology)	Therapy
Very low risk (<2 years, <550 g, stage I)	Nephrectomy alone
Stage I–II	Vincristine + dactinomycin
Stage III	Add doxorubicin + flank RT (radiotherapy)
Stage IV	Three drugs + lung RT (omitted if lung nodules clear by week 6)
Anaplasia; 1p/16q LOH (loss of heterozygosity)	Intensified regimens

- **Bilateral (stage V):** three-drug chemotherapy, then **NSS (nephron-sparing surgery) by week 12**.

Prognosis

- Favourable histology: survival >90%; late effects (cardiotoxicity, second cancers).

Diagram to draw: COG staging showing capsule, renal sinus and nodes.

> **Key points to remember:**
> - WT1 at 11p13 (WAGR, Denys-Drash); BWS at 11p15.
> - Any spill or positive node = stage III; sample ≥5 nodes.
> - COG operates first; SIOP gives chemotherapy first; bilateral → NSS by week 12.

Q3c. Hemodialysis access.

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 4, Q3b and 2025-Winter Paper 2, Q3c)

Introduction

- Haemodialysis needs blood flow of **250–350 mL/min**. Access failure is a major cause of morbidity in ESKD (end-stage kidney disease).

Types

Access	Construction	Ready	Comment
Radiocephalic AVF (arteriovenous fistula; Brescia–Cimino)	Radial artery to cephalic vein at wrist, end-to-side	6–8 weeks	**First choice**: best patency, fewest infections
Brachiocephalic AVF	At elbow	6–8 weeks	Higher flow, more steal
AVG (arteriovenous graft)	PTFE (polytetrafluoroethylene) loop or straight	2–3 weeks	More stenosis, thrombosis, infection
Tunnelled cuffed CVC (central venous catheter)	Right internal jugular	At once	Bridge or last resort; infection, central stenosis

- Non-tunnelled catheters: acute dialysis only.

Planning

- **Hierarchy**: AVF, then AVG, then CVC; **distal before proximal; non-dominant arm first**. KDOQI (Kidney Disease Outcomes Quality Initiative) 2019 favours an individual "ESKD Life-Plan".
- **Timing**: refer at eGFR (estimated glomerular filtration rate) <30; create the AVF about **6 months** before dialysis.
- **Vein preservation**: no venepuncture or PICC (peripherally inserted central catheter) in that arm; avoid subclavian lines.
- **Assessment**: pulses, **Allen test** (ulnar collateral flow), **Doppler vein mapping** (vessels about ≥2 mm).
- **Maturity "rule of 6s"**: flow >600 mL/min, diameter >6 mm, depth <6 mm, at 6 weeks.

Complications

Complication	Treatment
Failure to mature (juxta-anastomotic stenosis)	Angioplasty, ligate side branches
Stenosis (neointimal hyperplasia)	PTA (percutaneous transluminal angioplasty), stent graft
Thrombosis (usually on a stenosis)	Thrombectomy, then fix the stenosis
Steal syndrome (hand ischaemia)	Banding, DRIL (distal revascularisation–interval ligation)
Aneurysm; infection (CVC > AVG > AVF)	Repair; antibiotics, excise graft

- **Also know:** brachiobasilic transposition AVF; venous hypertension; high-output cardiac failure.

Monitoring

- Thrill and bruit, arm-elevation test, prolonged bleeding after needling, rising venous pressure; duplex when abnormal.

Recent advances

- **Endovascular AVF** (percutaneous creation); drug-coated balloons; bioengineered vessels.

Diagram to draw: radiocephalic end-to-side fistula at the wrist.

> **Key points to remember:**
> - AVF first, distal, non-dominant arm, 6 months ahead.
> - Rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
> - Thrombosis sits on a stenosis; steal → banding or DRIL.

Q3d. TRUS

Type: SAQ (10 marks) (also answered in 2017-Winter Paper 1, Q3e and 2015-Summer Paper 2, Q3f)

Read as: transrectal ultrasonography (TRUS) of the prostate.

Principle and technique

- The prostate lies directly in front of the rectum. Probes: end-fire, side-fire or biplane, **6–10 MHz** (higher frequency gives better resolution but less depth).
- Left lateral position after DRE (digital rectal examination); transverse and sagittal planes, base to apex.

Normal sonographic anatomy

Zone (McNeal)	Glandular share	TRUS appearance	Cancer share
Peripheral	~70%	Homogeneous, echogenic (posterior)	~70%
Central	~25%	Echogenic, at base	~5–10%
Transition	5–10% (BPH (benign prostatic hyperplasia))	Heterogeneous, anterior	~20–25%
Anterior fibromuscular stroma	Non-glandular	Hypoechoic	–

- TRUS separates echogenic posterior PZ (peripheral zone) from heterogeneous TZ (transition zone). **Corpora amylacea** mark the surgical capsule.
- Seminal vesicles 4.5–5.5 × 2 cm; absent seminal vesicle → ipsilateral renal agenesis in 79%.

Measurements

- **Volume (ellipsoid) = 0.52 × transverse × AP (anteroposterior) × length**; 1 mL ≈ 1 g; normal 20–25 g. **Planimetry** for brachytherapy.
- **PSAD (prostate-specific antigen density)** = PSA/volume; **>0.15** suggests cancer.

Findings

- **Cancer:** classically a hypoechoic PZ lesion, but **39% are isoechoic**; false positives are BPH nodules, prostatitis and infarcts. Capsular bulge suggests extracapsular extension.
- **Cysts:** Müllerian (midline), ejaculatory duct cyst (near midline), seminal vesicle cyst (lateral; renal agenesis).
- Abscess, calculi.

Indications

- **Guided biopsy:** 12-core systematic ± MRI (magnetic resonance imaging)-targeted fusion.
- Volume before BPH surgery, brachytherapy, cryotherapy; fiducial markers.
- **Azoospermia:** ejaculatory duct obstruction (seminal vesicle >1.5 cm wide, dilated ducts, midline cyst).
- Drainage of abscess or cysts; haematospermia.

Biopsy notes

- Periprostatic block. Complications: bleeding, **sepsis** (fluoroquinolone-resistant *E. coli*); the transperineal route lowers infection (EAU (European Association of Urology) preferred).

Advances

- Colour Doppler, contrast-enhanced TRUS, **elastography**; **micro-ultrasound (29 MHz)** with PRI-MUS scoring, detection comparable to mpMRI (multiparametric MRI).
- MRI-first pathway: PROMIS (2017): mpMRI sensitivity 93% vs 48% for TRUS biopsy.

Diagram to draw: transverse and sagittal TRUS views labelling PZ, TZ, seminal vesicles and urethra.

> **Key points to remember:**
> - 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15.
> - 39% of cancers isoechoic; hypoechoic PZ lesion is classic.
> - Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.

Q3e. Von Bruns Nest

Type: SAQ (10 marks)

Read as: von Brunn nests of the urothelium.

Definition

- **Von Brunn nests** are solid nests of benign urothelial cells lying in the **lamina propria**. They form when the basal layer of the surface urothelium buds downward and the buds pinch off.
- A common normal or reactive finding in most adult bladders, especially at the **trigone and bladder neck**; also in the ureter and renal pelvis.

Pathogenesis: the proliferative cystitis spectrum

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Chronic irritation (infection, stones, catheter, outlet obstruction)
   → urothelial buds → von Brunn nests (solid)
   → central degeneration and fluid → cystitis cystica (cysts)
   → glandular (columnar / goblet cell) metaplasia → cystitis glandularis
```

Lesion	Histology	Malignant potential
Von Brunn nests	Solid urothelial nests, lamina propria	None
Cystitis cystica	Nests with central cyst, urothelial lining	None
Cystitis glandularis (typical)	Cuboidal or columnar lining	Low
Cystitis glandularis, **intestinal type**	Goblet cells, mucin	Possible link to **adenocarcinoma**

Clinical features

- Usually incidental. When florid: irritative symptoms, haematuria, infection; cystoscopy shows submucosal nodules or cysts that can mimic tumour.
- **Associations:** chronic inflammation, outlet obstruction, **pelvic lipomatosis** (florid cystitis glandularis), bladder exstrophy.

Differential diagnosis (pathologist's trap)

- **Nested variant of urothelial carcinoma:** deceptively bland but aggressive (male:female 6:1). Clues: deep infiltrative nests reaching muscle, nuclear atypia.
- **Inverted papilloma:** endophytic nests with trabeculae, usually at the trigone.
- Nephrogenic adenoma; urothelial carcinoma with glandular differentiation.

Management

- None for isolated nests; treat the cause (infection, stone, catheter, obstruction).
- **TURBT (transurethral resection) or biopsy** if a mass is present, to exclude carcinoma.
- **Cystitis glandularis:** remove the irritant and arrange **endoscopic surveillance**, because rare change to adenocarcinoma is reported (especially the intestinal type and in exstrophy).
- Upper tract: benign, but coexisting upper-tract urothelial cancer cannot be excluded, so follow up.

Diagram to draw: urothelium budding into lamina propria, forming a solid nest, then a cyst, then glandular metaplasia.

Key points to remember:

- Benign urothelial nests in the lamina propria; precursor of cystitis cystica and glandularis.
- Only cystitis glandularis (intestinal type) carries adenocarcinoma risk → surveillance.
- The nested variant of urothelial carcinoma mimics von Brunn nests.

Q3f. Primary Hyperoxaluria

Type: SAQ (10 marks)

Definition

- Rare **autosomal recessive** errors of **hepatic glyoxylate metabolism**: glyoxylate is not converted to glycine and is oxidised to oxalate, an end product cleared only by the kidney.
- Hyperoxaluria = urinary oxalate >40 mg/day; in PH (primary hyperoxaluria) often **>100 mg/day**.

Types

	PH1 (~70%)	PH2 (~10%)	PH3 (~10%)
Enzyme (gene)	**AGT** (alanine-glyoxylate aminotransferase; **AGXT**), liver peroxisome, **pyridoxine (B6) cofactor**	GRHPR (glyoxylate reductase/hydroxypyruvate reductase)	HOGA1 (4-hydroxy-2-oxoglutarate aldolase), mitochondria
Urine marker	**Glycolate**	**L-glycerate**	4-hydroxy-2-oxoglutarate
Course	**Most severe**; ESKD (end-stage kidney disease) in 2nd–3rd decade	Milder, can reach ESKD	Childhood stones; rarely ESKD

Pathophysiology

- High urinary oxalate → calcium oxalate crystals in tubules → **nephrocalcinosis** and recurrent stones.
- **Vicious circle:** falling GFR (glomerular filtration rate) reduces oxalate clearance → plasma oxalate rises → **systemic oxalosis** (bone, retina, heart, vessels, skin).

Clinical features

- Spectrum from **infantile oxalosis with renal failure** to occasional adult stones; some diagnosed only after a transplant fails.
- Clues: childhood stones, nephrocalcinosis, family history, **calcium oxalate monohydrate** stones.

Diagnosis

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24-h urinary oxalate ↑ (child: spot oxalate:creatinine ratio)
   → urinary glycolate / L-glycerate / 4-hydroxy-2-oxoglutarate
   → genetic testing (AGXT, GRHPR, HOGA1)
   all normal → secondary (enteric, dietary) hyperoxaluria
```

- Plasma oxalate when GFR is low; screen siblings and related living donors.

Management

- **Hyperhydration** (about 3 L/m²/day), **potassium citrate**, normal calcium intake.
- **Pyridoxine** trial in all PH1: responsive mainly with the Gly170Arg mutation; it restores AGT activity and peroxisomal targeting.
- **Lumasiran** (RNA interference against glycolate oxidase, which cuts glyoxylate supply): **ILLUMINATE-A**: 84% vs 0% reached urinary oxalate ≤1.5 × upper limit of normal. **Also know:** nedosiran.
- Stones: ureteroscopy or PCNL (percutaneous nephrolithotomy); limit SWL (shock wave lithotripsy).

- **ESKD:** intensive dialysis, then **combined liver–kidney transplant for PH1** (only the liver replaces the enzyme; a lone kidney would be re-damaged). Kidney alone usually suffices for PH2.

Diagram to draw: hepatic glyoxylate pathway showing the AGT block (PH1) and the lumasiran target (glycolate oxidase).

> **Key points to remember:**
> - PH1: AGT (AGXT), pyridoxine cofactor, urinary glycolate; most severe.
> - Urinary oxalate often >100 mg/day; nephrocalcinosis; oxalosis when GFR falls.
> - Lumasiran (ILLUMINATE-A); combined liver–kidney transplant in PH1.

Rapid Revision Sheet – Practice Set 10 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Describe staging of CA prostate and management of non-organ confirmed CA prostate (LAQ)

- Seminal vesicle invasion is T3b; non-regional nodes are M1a.
- RT + long-term ADT (18–36 months) is the standard; ADT alone is inferior (SPCG-7).
- STAMPEDE: abiraterone 2 years for N1 or very high-risk M0 (metastasis-free survival HR 0.53).
- Early salvage RT preferred over adjuvant RT for most men after RP.

Q2. Classify CA bladder. Discuss in details organ preserving surgeries for TCC of bladder. (LAQ)

- WHO 2004/2016: PUNLMP, low grade, high grade; T3b = macroscopic perivesical fat.
- Re-TURBT within 6 weeks for T1, no muscle or incomplete resection.
- Partial cystectomy: solitary <3 cm, no CIS, frozen-section margins, always PLND.
- TMT: BC2001 MMC/5-FU 67% vs 48%; salvage cystectomy 10–30%.

Q3a. Allopurinol (SAQ)

- Xanthine oxidase inhibitor; active metabolite oxypurinol; 300 mg/day.
- Use: hyperuricosuric calcium oxalate stones (Ettinger 1986); uric acid stones need alkali first.
- Hypersensitivity 2–8%; HLA-B*58:01; cut azathioprine dose.

Q3b. Wilm's tumor (SAQ)

- WT1 at 11p13 (WAGR, Denys-Drash); BWS at 11p15.
- Any spill or positive node = stage III; sample ≥5 nodes.
- COG operates first; SIOP gives chemotherapy first; bilateral → NSS by week 12.

Q3c. Hemodialysis access. (SAQ)

- AVF first, distal, non-dominant arm, 6 months ahead.
- Rule of 6s: >600 mL/min, >6 mm, <6 mm deep, 6 weeks.
- Thrombosis sits on a stenosis; steal → banding or DRIL.

Q3d. TRUS (SAQ)

- 6–10 MHz; volume = 0.52 × T × AP × L; PSAD >0.15.
- 39% of cancers isoechoic; hypoechoic PZ lesion is classic.
- Seminal vesicle >1.5 cm wide: ejaculatory duct obstruction.

Q3e. Von Bruns Nest (SAQ)

- Benign urothelial nests in the lamina propria; precursor of cystitis cystica and glandularis.
- Only cystitis glandularis (intestinal type) carries adenocarcinoma risk → surveillance.
- The nested variant of urothelial carcinoma mimics von Brunn nests.

Q3f. Primary Hyperoxaluria (SAQ)

- PH1: AGT (AGXT), pyridoxine cofactor, urinary glycolate; most severe.
- Urinary oxalate often >100 mg/day; nephrocalcinosis; oxalosis when GFR falls.
- Lumasiran (ILLUMINATE-A); combined liver–kidney transplant in PH1.